

Si Yudong

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☐ Education

Tongji University

2019.09 - 2022.03

☐ Work Experience

AI Framework Engineer (Triton Compiler)@Intel.

2022.07 - 2025.07



Triton Compiler XPU Backend Development

• Feature Implementation

- Implemented global scratch memory addressing in backend launcher
- Built SPIR-V → Level Zero → SYCL AOT compilation pipeline, facilitating low-level issue reproduction.
- Created Intel specific Pass tritonintelgpu-rewrite-stack-ptr to optimize shared local memory (SLM) pointer handling.
- Integrated __spirv_RoundFToTF32INTEL SPIRV extension for enhanced FP32 precision in tl.dot DPAS lowering.

• Performance Optimization

- Established Triton-XPU's first benchmark system covering Softmax/GEMM/FA key kernels, with extensible support for vendor libraries (XeTLA/CUTLASS/oneDNN).
- Landing optimization passes like: Coalesce, AcceletateMatmul, RemoveLayoutConversion, Pipeline / Prefetch, Swizzling etc.
- Optimized critical kernels (e.g. GEMM) and achieved >90% performance of Intel's XeTLA library

• Bug Fixes

- Resolved 35+ High-priority compiler backend issues including: performance regressions, Layout propagation failures in IR passes, ut issues and so on.

PyTorch Ecosystem Optimization

• CI/CD Innovation

- Designed AWS Xeon-based Jenkins pipeline for PyTorch Inductor CPU Performance:
 - Automated collection of 200+ model performance metrics
 - Auto-generated performance reports

CUDA Test Engineer Intern@NVIDIA

2021.08 - 2021.11

CUDA Orin Simulator Practice

• Built CUDA safety & code coverage system for NVIDIA Orin t23x SOC

- Automated test platform using VDK virtual test suite
- Debugged failed/timeout test cases
- Automated test image version tracking

☐ Skills

• **Compiler:** Triton,LLVM,MLIR,SPIRV

• **Heterogeneous Computing:** oneAPI,OpenCL,CUDA,SYCL,Level Zero,ROCm

• **AI Framework:** Pytorch,IPEX

• **AI Infrastructure:** Jenkins,GitHub Actions,Docker,Vtune Profiler

• **Languages:** C/C++, Python, Bash